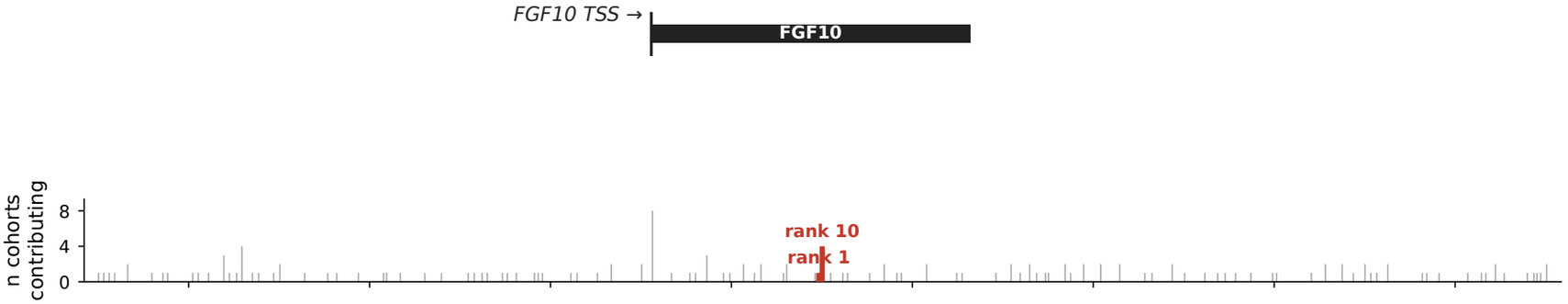
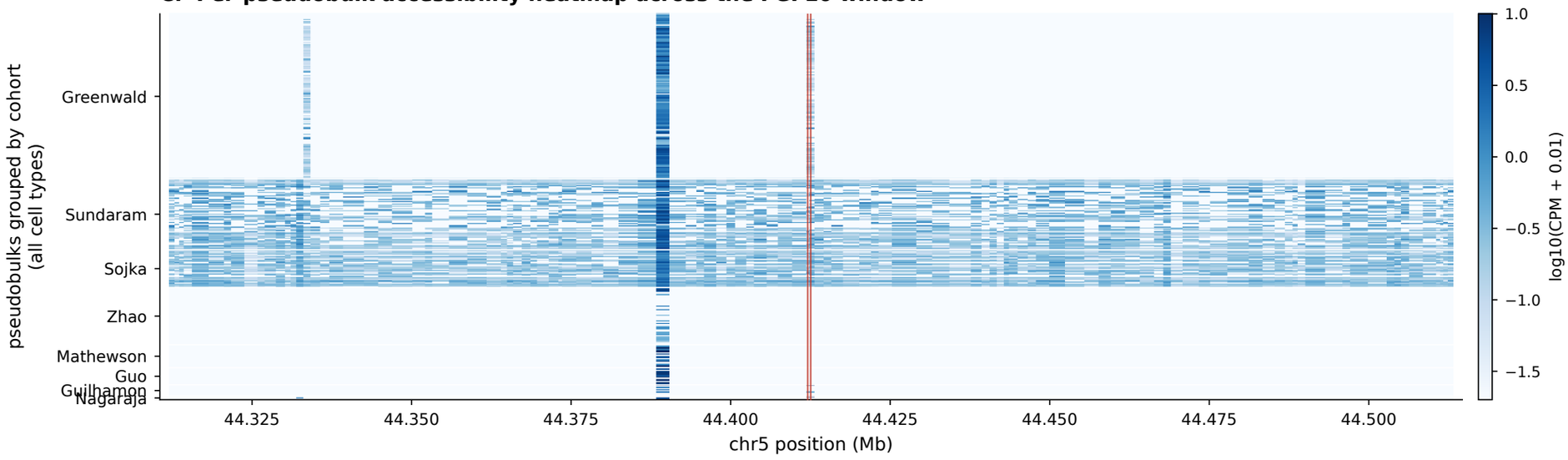
FGF10 is the pan-GBM enhancer candidate: cross-cohort breadth, patient-level heterogeneity

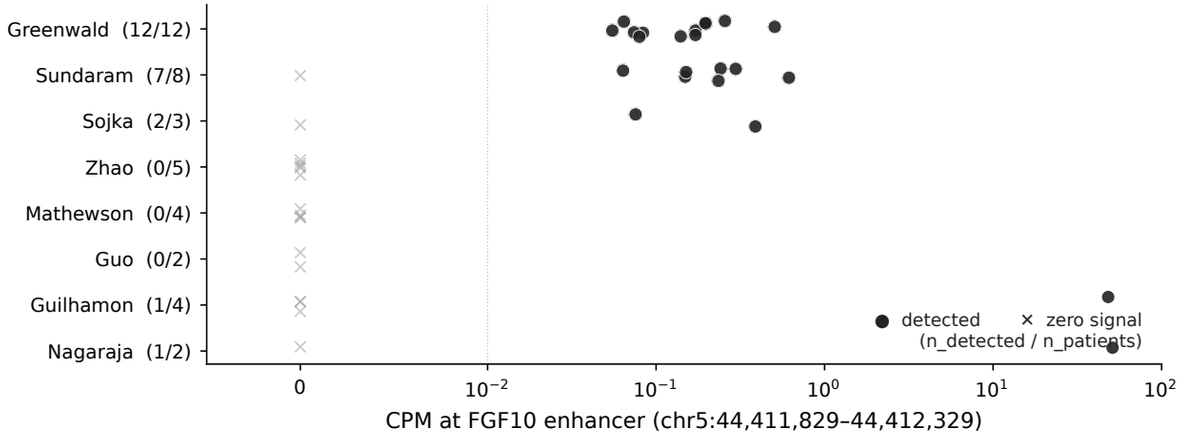
A. FGF10 locus (chr5p12) — pan-malignant enhancer candidate (rank 1 of pan_malignant top-100)



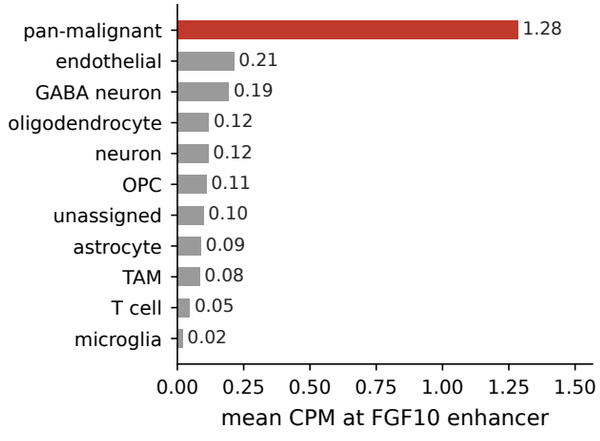
C. Per-pseudobulk accessibility heatmap across the FGF10 window



D. Per-patient malignant CPM at the FGF10 enhancer



E. Cell-type specificity



Scoring. peak = chr5:44,411,829–44,412,329 (−22 kb from FGF10 TSS, distal) · Z_arm_adjusted = 59.5 · Z_raw = 15.0 · target CPM = 2.62 · AAV safety ratio = 7.3

Replication. 23 patients across 5 cohorts · cohort_balance = 0.50 · greenwald_fraction = 0.07 · patient_fraction_chr7_elevated = 0.33 · arm chr5p (no known CNV)

Pattern. Broad low-level accessibility in Greenwald (12/12 patients, median 0.17 CPM), Sundaram (7/8), Sojka (2/3). Lone outlier patients in Guilhamon (1/4 at 48 CPM) and Nagaraja (1/2 at 51 CPM) carry most summed CPM. Guo, Mathewson, Zhao show zero signal. 23 of 48 patients detect any signal — real but heterogeneous.